Type MS · Scale a rate per 100 000 people to a whole population

Answer 1 of 5 · Question 3 part b.ii · 3 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part b.ii in this mark scheme extract

3 (a) (i)	arrow from top to bottom;		1
(ii)	• A is an artery, thicker muscle wall; • to pump blood from heart/to body; • B is a vein with a valve present; • to control direction of blood flow/to prevent backflow;	If no other marks scored award one for correct identification of both vessels	4
(iii)	capillary;		1
(b) (i)	• decrease in deaths; • in both sexes/males and females; • between 2005 - 2014/over whole range;		3
(ii)	• 60 000 000 ÷ 100 000 = 600; • 600 × 640; • = 384 000;		3

Total 12 marks

Type MS · Scale a rate per 100 000 people to a whole population

Answer 1 of 5 · Question 3 part b.ii · 3 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Scale a rate per 100 000 people to a whole population

Answer 2 of 5 · Question 6 part a.iii · 2 marks

2020 JAN · Paper 02 · find part a.iii in this mark scheme extract

Question number					
6	(a)	(i)	• more males than females die (from *S.aureus* and MRSA)(1) • number of deaths from both *S.aureus* and MRSA increase with age(1)	Allow other valid conclusions e.g more deaths from both microorganisms in males/females aged 85 and over	2
		(ii)	416.6 ÷ 1.8(1) 231.4:1 (1)	Allow 231:1 Allow full marks for correct final answer	2
		(iii)	41.2 + 21.9 = 63.1(1) 53 × 63.1 = 3344(1)	Allow 3344.3 Allow full marks for correct final answer	2
		(iv)	Any two from: • MRSA is resistant (1) • to (many) antibiotics(1) • (MRSA infection) difficult to treat(1) • reference to MRSA infection being (highly) contagious/out of control/easily transferred(1)		2
	(b)	(i)	mircroorganisms/bacteria/fungi	Ignore viruses Ignore named antibiotics	1
		(ii)	Any six from: • sterilise Petri dishes/agar/nutrient broth(1) • obtain sample(1) • sterile inoculating loop/sterile swab(1) • pass neck of flask containing culture through a flame(1) • smear culture/sample/swab onto agar(1) • reference to adding antibiotics (to agar)(1) • seal Petri dishes/cover with lid(1) • incubate(1) • measure area of clear zone around antibiotic discs(1)	Allow description of how loop/swab is sterilised	6

Total for Question 6 = 15 marks

Type MS · Scale a rate per 100 000 people to a whole population

Answer 3 of 5 · Question 4 part b · 3 marks

2021 JUN · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

number				
4	(a) (i)	A (bacterium); B doesn't cause it C doesn't cause it D doesn't cause it		1
	(ii)	• no of cases decreases as vaccination rate increases; • lag in effect; • fluctuations in number of cases; • despite percentage of vaccinations remaining constant;		4
	(b)	2.75 per 100 000; $\frac{2.75 \times 3\,450\,000;}{100\,000}$ = 95 people;	correct answer without working = 3 ECF = 2	3
	(c)	• inject weakened/attenuated organism; • causes antibody production; • antibodies remain in blood; • memory cells formed; • on infection antibodies can respond quickly;		5

Total 13 marks

Type MS · Scale a rate per 100 000 people to a whole population

Answer 3 of 5 · Question 4 part b · 3 marks

2021 JUN · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Scale a rate per 100 000 people to a whole population

Answer 4 of 5 · Question 6 part b.ii · 3 marks

2021 NOV · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

number				
6 (a)	(i)	• both increase; • males more than females; • fluctuations; accept any two from;		3 marks
	(ii)	• layer of actively dividing cells; • can become uncontrolled; • when exposed to ultraviolet/sun light;		2 marks
(b)	(i)	17 - 48; = 31 per 100 000; $\frac{31}{25}$ = 1.2/1.24 per 100 000;	allow 16.7 - 17 and 47.8 - 48 then ECF	3 marks
	(ii)	20 per 100 000; $\frac{20 \times 7\,000\,000;}{100\,000}$ = 1400;	ECF	3 marks

Total 11 marks

Type MS · Scale a rate per 100 000 people to a whole population

Answer 4 of 5 · Question 6 part b.ii · 3 marks

2021 NOV · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Scale a rate per 100 000 people to a whole population

Answer 5 of 5 · Question 4 part c.ii · 3 marks

2024 NOV · Paper 02 · extract 1 of 2 · find part c.ii in this mark scheme extract

4 (a)	• diarrhoea by clean water/safe water/ bottled water/boiled water/ sewage treatment/chlorination/good sanitation/not mixing sewage with drinking water/washing hands/good hygiene/ cooking food thoroughly; • malaria by nets/draining swamps/insecticides/ insect repellent's/wearing clothing to cover arms (and legs/mosquito coils/anti-malarial/vaccination ;	Ignore ORT	2
(b) (i)	antibiotics;		1
(ii)	• elderly have weakened/lowered immune system/immunocompromised; • indoor spaces makes transmission easier/reduced air flow/ reduced ventilation/build- up of air droplets with pathogens/ people closer together;		2
(c) (i)	• diet; • exercise; • smoking: • drinking alcohol: • stress	Accept drugs	2
(ii)	• $\frac{5\,000\,000}{100\,000} = 50$; • 50 × 445 = 22 250; • 22 300; ***22,250 scores 2 marks*** ***22,300 scores 3 marks***	***1 mark for ×445 if there are no other marks***	3
(iii)	• use of 445 and 185; • $\frac{445}{185}$; • 2.41:1/2.405:!/2.4:1= 3 marks;	ecf from mp1 **if they have done the correct calculation and 2:1 is on the answer line they score** 2 marks	3
(iv)	• people live longer/elderly population; • these diseases are more common in older people; • less likely to die younger from infectious diseases/better medical facilities/treatment; • different diet/ more meat/saturated fats/processed food; • drinking more alcohol;	People live longer so there is more cancer and dementia/these diseases= 2 marks	2

Total 15 marks

Type MS · Scale a rate per 100 000 people to a whole population

Answer 5 of 5 · Question 4 part c.ii · 3 marks

2024 NOV · Paper 02 · extract 2 of 2 · find part c.ii in this mark scheme extract

Question number	Answer	Notes	Marks